

DARPA-PA-26-05 MATHBAC

Proposal Instructions and Volume I Template

*Use of this template is **mandatory** for all proposal submissions to this PA. Proposals not meeting the format prescribed herein may not be reviewed.*

This document must include all components described herein and must be submitted in PDF or Microsoft Word formats.

All submissions must be written in English, and all pages shall be formatted for printing on 8-1/2 by 11-inch paper with 1-inch margins and font size no smaller than 12-point. Font sizes of 8 or 10-point may be used for figures, tables, and charts.

Proposers are encouraged to submit concise but descriptive proposals. Specific examples of problems, approaches, or goals are preferred to qualitative generalities. The Government will not consider pages in excess of the page count limitations, as described herein. Proposals with fewer than the maximum number of pages will not be penalized. Additional information not explicitly called for in the Technical and Management Volume must not be submitted with the proposal but may be included as links in the bibliography. Such materials will be considered for the reviewers' convenience only and not evaluated as part of the proposal.

Technical and Management Volumes shall not exceed a maximum of **20** pages.

Page limit includes:	Page limit does NOT include:
Sections 2-5	Cover Sheet
Technical figures, tables, charts	Official transmittal letter (Section 1)
	Table of Contents
	Sections 6-13
	Resumes (Optional)
	References/Bibliography (Optional)
	Technical Papers (Optional, no more than 3)

Proposals must be submitted per the instructions outlined herein *and received by DARPA* no later than the due date and time listed in the Overview Information section. Proposals received after this time and date may not be reviewed. Please visit [Proposer Instructions and General Terms and Conditions](#) for instructions on how to submit your proposal through the Broad Agency Announcement Tool (BAAT).

COVER SHEET
[PRIME ORGANIZATION LOGO]

Proposal Title	SCBE Mathematical Framework for Agentic Communication Protocols
Proposer Organization	ISSAC D DAVIS / SCBE-AETHERMOORE
Topic Area (TA)	TA1
Type of Organization	Other Small Business (sole proprietorship, minority-owned)
Proposer Reference Number, if any	N/A
Technical Point of Contact (POC)	Name: Issac Davis Address: 2361 E 5th Ave, Port Angeles, WA 98362-9014 Telephone: (360) 808-0876 Email: issdandavis7795@gmail.com
Administrative POC	Name: Issac Davis Address: 2361 E 5th Ave, Port Angeles, WA 98362-9014 Telephone: (360) 808-0876 Email: issdandavis7795@gmail.com
Place(s) of Performance	2361 E 5th Ave, Port Angeles, WA 98362-9014 (remote / local)
Period(s) of Performance	16 months from award date
Months	16
Other Team Members (subawardees and consultants), if any	Technical POC Name: Collin Hoag Organization: Hoags Inc. Organization Type: Supporting Subcontractor (bounded scope)
Total Price of Phase I	Year 1: \$[fill when award date / FY boundary known] Year 2: \$[remainder] Year 3: N/A Total: \$839,000
SAM.gov Unique Entity ID (UEI)ⁱ	J4NXHM6N5F59
Taxpayer identification number (TIN)ⁱⁱ	[Enter TIN directly in Word — do not record here]
Commercial and Government Entity (CAGE) codeⁱⁱⁱ	1EXD5

Administration Office POC at Defense Contract Management Agency (DCMA)^{iv} or Office of Naval Research (ONR)^v, if known	Name: Unknown at proposal stage Address: — Telephone: —
Audit Office POC at Defense Contract Audit Agency (DCAA)^{vi}, if known	Name: Unknown at proposal stage Address: — Telephone: —
Date Proposal was Prepared	2026-06-[fill on submission day]
AI-Generated Material, if any	Yes. Claude (Anthropic) was used to assist in drafting and formatting sections of this proposal. All technical content, mathematical derivations, benchmark evidence, and strategic decisions are authored by Issac Davis / SCBE-AETHERMOORE.
Proposal Validity Period (minimum 120 days)	120 days from submission

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1. Official Transmittal Letter

Issac Davis / SCBE-AETHERMOORE

2361 E 5th Ave, Port Angeles, WA 98362-9014

issdandavis7795@gmail.com | (360) 808-0876

DARPA / MATHBAC Program

DARPA-PA-26-05 | Full Proposal Reference: DARPA-PA-26-05-MATHBAC-PA-010

I am pleased to submit this full proposal in response to DARPA-PA-26-05 (MATHBAC). SCBE-AETHERMOORE proposes a 16-month Phase I effort at \$839,000 to develop a formal mathematical framework for governing multi-agent AI communication dynamics using hyperbolic geometry. All work will be performed at the prime organization under my direction as Principal Investigator.

Issac Davis, Principal Investigator

ISSAC D DAVIS / SCBE-AETHERMOORE | UEI J4NXHM6N5F59 | CAGE 1EXD5

2. Proposal Summary

SCBE-AETHERMOORE proposes to develop a composed-operator governance framework $T = L_{14} \circ \dots \circ L_1$ for multi-agent AI communication in scientific discovery tasks. The framework maps agent communication acts to points in a Poincaré ball (hyperbolic space) and measures safety by hyperbolic distance from the safe-operation origin. Safety cost scales exponentially with adversarial drift, making attacks computationally infeasible.

Current state-of-the-art multi-agent AI safety relies on post-hoc filtering, red-teaming, and RLHF fine-tuning. None of these approaches provide a formal mathematical characterization of when a multi-agent collective is communicating correctly versus drifting toward misalignment. Spera (arXiv:2603.15973, 2026) proves that 42.6% of 900 real multi-agent trajectories have conjunctive safety dependencies that flat classifiers cannot detect.

Success will enable DARPA to field mathematically-auditable multi-agent AI systems with verifiable communication integrity across scientific subdomains (NMR spectroscopy, cheminformatics, and Phase II: materials discovery and theorem-proving).

Cost: \$839,000 | Duration: 16 months | Phase II ROM: ~\$382,000.

The key innovation is governance as geometry. Five physical axioms (Unitarity, Locality, Causality, Symmetry, Composition) are enforced layer-by-layer; the composition is the governance certificate. Pre-submission benchmark evidence: 13/13 terminal-bench parity, 172/173 Petri adversarial classification (99.42%), 5/5 real-patch task harness wins.

- What is the proposed work attempting to accomplish or do?
- How is the work performed today (what is the state of the art or practice), and what are the limitations?
- Who will care, and what will be the impact if the work is successful?
- How much will it cost, and how long will it take?
- What is new in your approach, and why do you think it will be successful?

The summary should include a description of the key technical challenges, a concise review of the technologies proposed to overcome these challenges and achieve the project's goal, and a clear statement of the novelty and uniqueness of the proposed work.]

3. Goals and Impact

Goal: Develop and empirically validate a formal mathematical framework for governing multi-agent AI communication dynamics. The framework treats governance as a geometric property of the collective's communication trajectory in Poincaré ball space, not as a post-hoc classifier applied to individual messages.

Phase I targets: (1) Lyapunov convergence constants η , b per subdomain (M3). (2) CDPTI $\geq 95\%$ recall, $\leq 5\%$ FPR vs. Mixtral-8x7B baseline (M13). (3) Four proposer-added metrics (MEE, ACV, CDPTI, PIS) computable from IV&V bundle at every milestone. (4) NMR validation: Task A (1H NMR $\rightarrow$ structure) + Task B (2D COSY), Karplus rediscovery demonstrated through governed agent communication. (5) Subdomain agnosticism demonstrated on cheminformatics secondary lane (M9).

Impact if successful: Scientifically reliable multi-agent AI systems with verifiable communication integrity. The composed-operator framework provides a common mathematical substrate for TA1 and feeds directly into TA2's principle-extraction mandate. The NMR Karplus rediscovery experiment is a falsifiable, graded demonstration that governed multi-agent collectives can recover scientific principles through communication alone — without the principle appearing in any training context.

4. Technical Plan

TECHNICAL APPROACH: COMPOSED OPERATOR GOVERNANCE

The 14-layer pipeline $T = L_{14} \circ \dots \circ L_1$ implements five physical axioms across layers: Unitarity (L2, L4, L7), Locality (L3, L8), Causality (L6, L11, L13), Symmetry (L5, L9, L10, L12), Composition (L1, L14). Each agent A_i is modeled as $\Phi_i : T_i^* \rightarrow B^n_i$, mapping token strings to points in the Poincaré ball via realification, LWS phi-weighting, and the radial exponential map.

MC-1 (Agent Operator Model): One-step latent operator $S_i = \text{Phase}_i \circ \text{Breathe}_i \circ \text{Move}_i$. Agent interactions scheduled by Juggling Scheduler (physics-based, 7 rules). Three interaction modes: output-only (b1), latent-access (b2), mixed (b3).

MC-2 (Protocol Graph): Trust-weighted directed Laplacian $L_H(G_t)$ over communication graph G_t . Edge weights $w_{ij}(t) = \text{sign}(H_{ij} - H_{\min}) \cdot \|\log(S_{ij})\|$. $\text{CDPTI} = \text{Re}(\lambda_2(L_H(G_t))) / \text{Re}(\lambda_2(L_H(G_{t_0})))$. Cascade (monotone DAG) and recurrence ($\text{SCC} \geq 3$ agents) detectors trigger ESCALATE.

MC-3 (Harmonic Wall + Lyapunov): $H(d, pd) = 1 / (1 + \phi \cdot d_H + 2 \cdot pd) \in (0, 1]$. Lyapunov candidate $V(t) = -\log H(d_H(\psi(t), \psi_{\text{safe}}), pd(t))$. Convergence: $V(t+1) \leq (1-\eta)V(t) + b$. Constants η, b are M3 empirical deliverables.

MC-4 (Protocol Optimization): Bicriteria MDP: minimize $E_\pi[\sum_t c_t]$ s.t. $\Pr_\pi[\text{success}] \geq \tau$ and $\text{ACV}(\pi) \geq \text{ACV}_{\min}$. Lagrangian relaxation with subgradient updates. Step-cost vector $c_k = (\text{token}, \text{time}, \text{failure}, \text{energy}, \text{compliance}) \in \mathbb{R}_{\geq 0}$.

MC-5 (Oracle): Triplet condition — $MI \geq I_{\min}$ AND spectral coherence $\geq \gamma_{\min}$ AND $MEE \geq MEE_{\min}$. All three thresholds are M3 calibration deliverables. Oracle is falsifiable: ablate any one check, true-positive rate drops measurably.

Science Subdomain (TA1-(3)): Primary SSM ChemBERTa-77M (seyonec/ChemBERTa-zinc-base-v1, 768-dim, probe layer 10/12). Orchestrator Qwen2.5-Coder-0.5B-Instruct. Task Family A: 1H NMR $\rightarrow$ structure (Karplus equation $J = A \cdot \cos^2\theta + B \cdot \cos\theta + C$). Task Family B: 2D COSY cross-peak assignment. Validation lane: Hammett σ - ρ from reaction corpus.

Risk mitigations: R1 (ChemBERTa probe noise) — fallback to output-logit operator. R2 (Mixtral outperforms raw accuracy) — SCBE value is governance-overhead-adjusted parity, not raw supremacy. R3 (IV&V challenge problem scope mismatch) — FAQ Q1 filed 2026-05-31; secondary lane + Phase II plan reduce exposure. Full risk register in §2.3 of this volume.

5. Capabilities

SCBE-AETHERMOORE is a sole proprietorship (UEI J4NXHM6N5F59, CAGE 1EXD5, SAM.gov Active, minority-owned small business) built by PI Issac Davis. The organization has independently developed the SCBE 14-layer pipeline (62+ TypeScript modules, 40+ Python modules), the HYDRA multi-agent orchestration system, and the Juggling Scheduler physics-based task-flight coordinator.

Benchmark evidence (all pre-submission):

- 13/13 neutral-task oracle parity on terminal-bench-core-0.1.1 (commit f3fb4aa3c, zero governance overhead)
- 172/173 adversarial seeds correctly classified on Anthropic Petri benchmark (99.42%; 0.58% false-allow after regex v7, 2026-05-08)
- L13 RuntimeGate fast-path: 240 cases, p95 = 0.1095ms (target: <100ms, PASS)
- Real-patch task harness: 5/5 SCBE wins vs. 0/5 baseline on deterministic repair fixtures
- Semantic sphere benchmark: 26/26 surfaces, 505 iterations (2026-05-30T01:41:06Z)
- Bijective Sacred-Tongue encoder: 25/25 (100%) gate on deterministic test suite (commit 6538f4db)

Prior art (publicly timestamped): The Six Tongues Protocol (KDP ASIN B0GSSFQD9G, 2025); USPTO Provisional Patent Application 63/961,403. Mathematical foundations independently derived: Lyapunov stability, CBF, port-Hamiltonian operator structure, Poincaré ball metric (Khalil; Marsden/Ratiu; Cannon).

Prior government contracts: None. No OCI.

6. Management Plan

Prime: ISSAC D DAVIS / SCBE-AETHERMOORE. PI Issac Davis holds sole technical and management authority.

Supporting Subcontractor: Hoags Inc. (Collin Hoag; UEI DUHWVUXFNPV5 / CAGE 15XV5). Hoags Inc. provides DAVA bare-metal background-IP corroborating support, bounded to §5.1.6/§11.3 of the Phase I

SOW and Annex A Part 2. Hoags is not PI or co-PI. Phase I is fully executable by SCBE without Hoags support.

Milestone payment schedule (7 milestones, triggered by observable technical events):

M1 \$84,000 (Mo. 1) M2 \$109,000 (Mo. 3) M3 \$151,000 (Mo. 6)

M4 \$151,000 (Mo. 9) M5 \$151,000 (Mo. 13) M6 \$84,000 (Mo. 14) M7 \$109,000 (Mo. 16)

Risk register summary: R1 ChemBERTa probe noise (med/med); R2 Mixtral outperforms raw accuracy (low/med); R3 IV&V scope mismatch (low/high); R4 MC-2 instrumentation delay (med/med); R5 \$2M cap (med/high — addressed by \$839K lean budget); R6 BAAT upload deadline (med/critical); R7 prior publication IP (low/med); R8 teaming agreement execution (med/med).

Prime		
Issac Davis	ISSAC D DAVIS / SCBE-AETHERMOORE	Non-U.S. Organization: No Non-U.S. Individual: No FFRDC: No
		Non-U.S. Organization: No Non-U.S. Individual: No FFRDC: No
Subawardees/Consultants		
Collin Hoag	Hoags Inc.	Non-U.S. Organization: No Non-U.S. Individual: No FFRDC: No

7. Personnel, Qualifications, and Commitments

PI: Issac Davis | ISSAC D DAVIS / SCBE-AETHERMOORE | 100% effort Phase I

Qualifications:

- Sole developer of SCBE-AETHERMOORE AI safety/governance framework (62+ TypeScript + 40+ Python modules)

- Independent derivation: Lyapunov stability analysis, control-barrier functions, port-Hamiltonian operator structure, Poincaré ball metric
- HuggingFace: issdandavis (multiple trained/deployed models including phdm-21d-embedding, spiralverse-ai-federated-v1)
- Bijective Sacred-Tongue encoder: 25/25 gate; Petri adversarial: 172/173 (99.42%)
- Prior art: KDP ASIN B0GSSFQD9G; USPTO Provisional 63/961,403
- SAM.gov Active; UEI J4NXHM6N5F59; CAGE 1EXD5; sole prop; minority-owned
- No prior Government contracts; no OCI

Subcontractor: Collin Hoag / Hoags Inc. (bounded supporting scope, DAVA background IP). Not a key person for Phase I technical deliverables.

8. Organizational Conflict of Interest Affirmations and Disclosure

10.a. Are any proposed team members currently receiving support from DARPA under a separate award? ☐ No (X) ☐ Yes

Answer: No. Neither Issac Davis nor Hoags Inc. is receiving DARPA support under a separate award.

10.b. Did any proposed team member participate in developing the requirements for this solicitation? ☐ No (X) ☐ Yes

Answer: No.

10.c. Are there other potential OCI concerns? ☐ No (X) ☐ Yes

Answer: No. SCBE-AETHERMOORE has no prior Government contracts and is a sole proprietorship with no overlapping government work that could create a conflict of interest.

- a. Are any of the proposed individual team members *or* their respective organizations (whether prime or subawardee or consultant) *currently* providing SETA, A&AS or similar

support to DARPA? ☐ No ☐ Yes

- b. Did any of the proposed individual team members *or* their respective organizations (whether prime or subawardee or consultant) provide SETA, A&AS or similar support to DARPA within one calendar year of this proposal submission? ☐ No ☐ Yes

[If you answered "Yes" to 10.a OR 10.b, provide the following information for each applicable team member:

- The name of the DARPA office receiving the support;
 - The prime contract number;
 - Identification of proposed team member (subawardee, consultant) providing the support; and
 - An OCI mitigation plan in accordance with FAR 9.5.]
- c. Are there any other potential Organizational Conflicts of Interest involving any of the proposed individual team members *or* their respective organizations (whether prime or subawardee or consultant)? ☐ No ☐ Yes [If yes, provide the following information for each applicable team member:
- Identification of applicable team member; and
 - An OCI mitigation plan in accordance with FAR 9.5.]

9. Novelty of Proposed Work

Has the proposed work been submitted to any other Government solicitation? ☐ No (X) ☐ Yes

Has the proposed work already received funding or a positive funding decision? ☐ No (X) ☐ Yes ☐ Decision pending

The composed-operator governance framework is novel and has not been submitted to any other Government solicitation. Prior public disclosures (KDP ASIN B0GSSFQD9G; USPTO Provisional 63/961,403) establish timestamped prior art but do not represent Government funding.

" No " Yes [If yes, provide the following information]

- Solicitation number _____
- Agency _____
- Proposed work has already received funding or a positive funding decision.

" No " Yes " Decision pending]

10. Intellectual Property (IP)

Technical Data and Computer Software:

IP restrictions: Yes. The following are asserted as Background IP (Government Purpose Rights proposed):

1. The Six Tongues Protocol — KDP ASIN B0GSSFQD9G (2025). Prior published work establishing the Langues Weighting System and Sacred Tongue phi-metric basis.
2. USPTO Provisional Patent Application 63/961,403 — SCBE composed-operator governance framework. Filing date establishes priority for the 14-layer pipeline architecture.
3. SCBE-AETHERMOORE open-source codebase (github.com/issdandavis/SCBE-AETHERMOORE). Government Purpose Rights are proposed for all Phase I deliverables developed under this agreement.

Commercial software: No commercial software deliverables are proposed.

Patents: USPTO Provisional Application 63/961,403 is owned by Issac Davis / SCBE-AETHERMOORE. No other patents encumber the proposed effort. FAQ Q3 filed 2026-05-31 to confirm DARPA treatment of prior published work.

a. Technical Data and Computer Software

Are you asserting any IP restrictions on any technical data or non-commercial computer software that will be delivered to the Government? ☐ No ☐ Yes

[If yes, list all proprietary claims to results, prototypes, deliverables or systems supporting and/or necessary for the use of the proposed research, results, prototypes and/or deliverables. Provide a short summary for each item asserted with less than unlimited rights that describes the nature of the restriction and the intended use of the intellectual property in the conduct of the proposed research. Use the following format for these lists.]

NONCOMMERCIAL				
Technical Data and/or Computer Software To be Delivered with Restrictions	Summary of Intended Use in the Conduct of the Research	Basis for Assertion	Asserted Rights Category	Name of Person Asserting Restrictions

Are you proposing any commercial software that will be delivered to the Government? ☐ No ☐ Yes

[If yes, please provide licensing terms.]

COMMERCIAL				
Technical Data and/or Computer Software To be Delivered with Restrictions	Summary of Intended Use in the Conduct of the Research	Basis for Assertion	Asserted Rights Category	Name of Person Asserting Restrictions

b. Patents

Does the proposed effort involve using patented inventions that are owned by or assigned to the proposing organization or individual? ☐ No ☐ Yes [If yes, provide documentation proving ownership or possession of appropriate licensing rights to all patented inventions to be used for the proposed project. If a patent application has been filed for an invention, but it includes proprietary information and is not publicly available, provide documentation that includes: the patent number, inventor name(s), assignee names (if any), filing date, filing date of any related provisional application, and summary of the patent title, with either: (1) a representation of invention ownership; or (2) proof of possession of appropriate licensing rights in the invention (i.e., an agreement from the owner of the patent granting license to the proposer).]

11. Bibliography (Optional)

1. Khalil, H.K. (2002). Nonlinear Systems (3rd ed.). Prentice Hall. [Lyapunov Theorem 4.16; CBF]
2. Marsden, J.E. and Ratiu, T.S. (1999). Introduction to Mechanics and Symmetry. Springer. [Port-Hamiltonian structure]
3. Cannon, J.W. et al. (1997). Hyperbolic Geometry. MSRI. [Poincaré ball model]
4. Poppi, G. et al. (2025). Hyperbolic Safety-Aware Vision-Language Models. CVPR 2025. arXiv:2503.12127.
5. Cinà, G. et al. (2026). Harnessing Hyperbolic Geometry for Harmful Prompt Detection. arXiv:2604.06285.
6. Spera, S. (2026). Safety is Non-Compositional: A Formal Framework for Capability-Based AI Systems. arXiv:2603.15973.
7. Bhardwaj, A. (2026). Agent Behavioral Contracts. arXiv:2602.22302.
8. Anthropic. Petri adversarial benchmark (173 seeds, 2026). Internal reference; results documented in SCBE-AETHERMOORE repo.
9. terminal-bench-core-0.1.1. Commit f3fb4aa3c. SCBE-AETHERMOORE governance evidence brief.
10. BMRB (Biological Magnetic Resonance Data Bank). <https://bmr.io/>

11. SDBS (Spectral Database for Organic Structure Compounds). <https://sdb.sdb.aist.go.jp/>
12. seyonec/ChemBERTa-zinc-base-v1. HuggingFace.
13. Qwen/Qwen2.5-Coder-0.5B-Instruct. HuggingFace.
14. mistralai/Mixtral-8x7B-Instruct-v0.1. HuggingFace.

ⁱ The SAM-generated UEI becomes the official identifier for doing business with the U.S. Government. This replaces DUNS.

ⁱⁱ See <https://www.irs.gov/individuals/international-taxpayers/taxpayer-identification-numbers-tin> for information on requesting a TIN. Note, requests may take from 1 business day to 1 month depending on the method (online, fax, mail).

ⁱⁱⁱ A CAGE Code identifies companies doing or wishing to do business with the Federal Government.

^{iv} <https://pubapp.dcmi.mil/CASD/CasdSearch.do>.

^v <http://www.onr.navy.mil/Contracts-Grants/Regional-Contacts.aspx>.

^{vi} <http://www.dcaa.mil/Home/Locator?title=Locator>